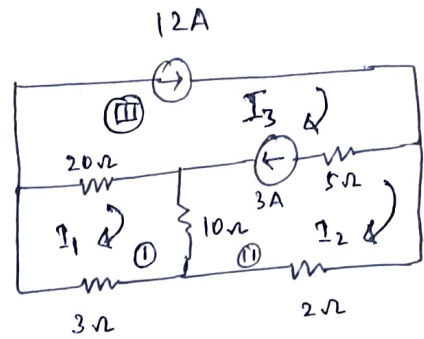


Q.1.

Assuming I_1 , I_2 & I_3
current in loop ①, ② & ③



$$\text{So, } I_3 = 12 \text{ A}$$

$$I_3 - I_2 = 3 \text{ A} \Rightarrow I_2 = 9 \text{ A}$$

Now, applying KVL in loop ①

$$20(I_1 - I_3) + 10(I_1 - I_2) + 3I_1 = 0$$

$$20(I_1 - 12) + 10(I_1 - I_2) + 3I_1 = 0$$

$$20(I_1 - 12) + 10(I_1 - 9) + 3I_1 = 0$$

$$33I_1 = 330$$

$$I_1 = 10 \text{ A}$$

current through 20Ω resistor is $I_1 - I_3 = -2 \text{ A}$

So,

$$V_{20\Omega} = 20(-2) \text{ V} = -40 \text{ V}$$

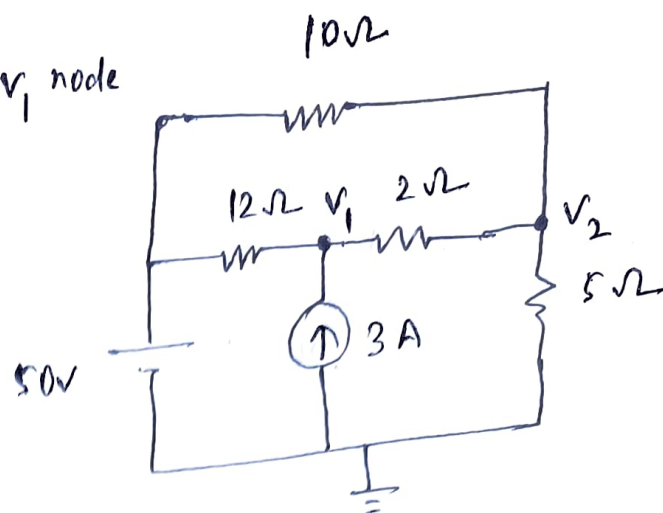
Q.2

Applying Nodal analysis at V_1 node

$$-3 + \frac{V_1 - V_2}{2} + \frac{V_1 - 50}{12} = 0$$

on simplifying

$$7V_1 - 6V_2 = 86 \quad \text{--- (1)}$$



Nodal analysis at node V_2

$$\frac{V_2}{5} + \frac{V_2 - 50}{10} + \frac{V_2 - V_1}{2} = 0$$

on simplifying

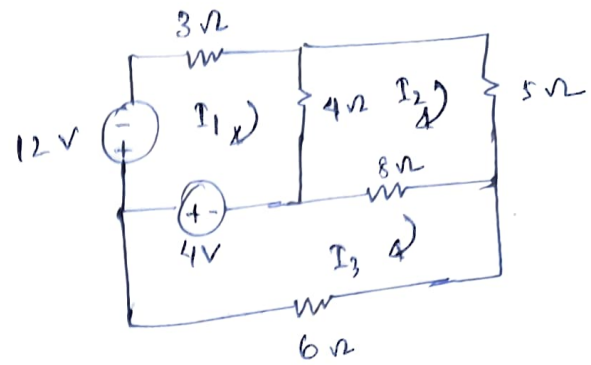
$$-5V_1 + 8V_2 = 50 \quad \text{--- (2)}$$

solving eqⁿ (1) & eqⁿ (2), we get

$$V_1 = 38V, \quad V_2 = 30V$$

so, current through 5Ω resistor is $\frac{V_2}{5} = \frac{30}{5} A = 6A$

3



Applying KVL in loop 1,

$$3I_1 + 4(I_1 - I_2) - 4 + 12 = 0$$

$$\Rightarrow 7I_1 - 4I_2 = -8 \quad \text{--- (1)}$$

$$\text{KVL in loop 2} \Rightarrow 5I_2 + 8(I_2 - I_3) + 4(I_2 - I_1) = 0$$

$$\Rightarrow -4I_1 + 17I_2 - 8I_3 = 0 \quad \text{--- (2)}$$

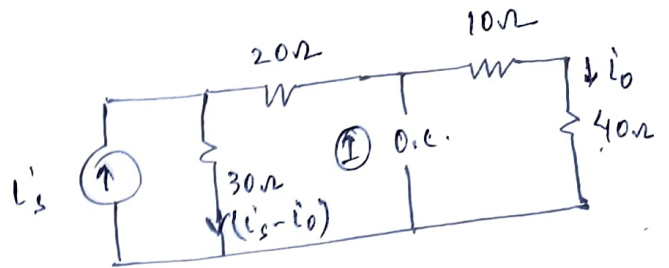
$$\text{KVL in loop 3} \Rightarrow 6I_3 + 4 + 8(I_3 - I_2) = 0$$

$$\Rightarrow 0I_1 - 8I_2 + 14I_3 = -4 \quad \text{--- (3)}$$

solving eqⁿ (i), (ii) & (iii)

$$I_1 = -1.53 \text{ A}, \quad I_2 = -0.676 \text{ A}, \quad I_3 = -0.672 \text{ A}$$

Q.4



A/q,

When no current flows through 10V source, it can be treated as open circuit

so, according to KCL, current through 30Ω resistor will be $(i_s - i_o)$

so, Applying KVL in loop ①

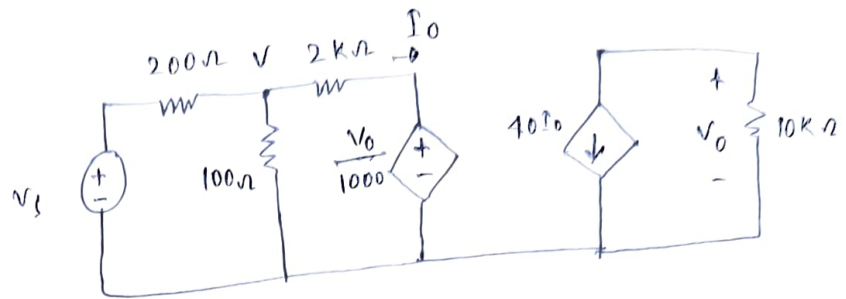
$$20i_o + 10i_o + 40i_o + 30(i_s - i_o) = 0$$

$$70i_o = 30(i_s - i_o)$$

$$100i_o = 30i_s$$

$$\frac{i_o}{i_s} = \frac{3}{10}$$

5



So, $V_0 = -40 I_0 (10k\Omega)$ — (i)

Also,

$$\frac{V - V_s}{200} + \frac{V}{100} + \frac{V - \frac{V_0}{1000}}{2000} = 0$$

on simplifying

So, $-10V_s - \frac{V_0}{1000} + 31V = 0$ — (ii)

Also,

$$I_0 = \frac{V - \frac{V_0}{1000}}{2000}$$
 — (iii)

So, ~~the~~ on simplifying

$$V = 2000 I_0 + \frac{V_0}{1000}$$
 — (iv)

carrying V from eqⁿ (iv) into eqⁿ (ii)

$$-10V_s - \frac{V_0}{1000} + 31 \left(2000 I_0 + \frac{V_0}{1000} \right) = 0$$
 — (v)

Putting $I_0 = -\frac{V_0}{400000}$ from eqⁿ (i) into eqⁿ (v)

we get $\boxed{\frac{V_0}{V_s} = -80}$

Q. 6 a) V_a & V_b are dependent variable i.e. $V_b - V_a = 16V$.
We can not express the current in branch between V_a & V_b .

b) Applying nodal analysis at node V_a

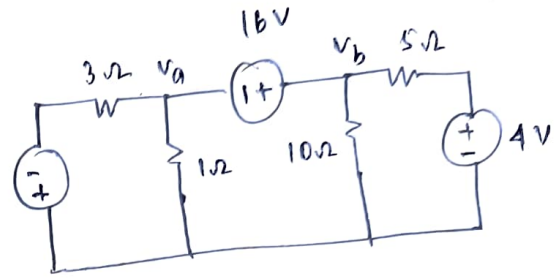
$$\frac{V_a + 6}{3} + \frac{V_a}{1} + \frac{V_b}{10} + \frac{V_b - 4}{5} = 0 \quad \text{--- (i)}$$

$$\text{and } V_b - V_a = 16 \quad \text{--- (ii)}$$

Solving Eqⁿ (i) & (ii)

$$V_a = -3.77V$$

$$V_b = 12.23V$$



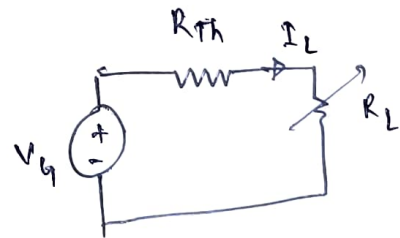
Q. 7. (a) $V_G = V_{TH} = V_{OC}$
 $R_{TH} = R_L$

$$\text{So, } P_L = I_L^2 R_L$$

$$P_L = \left(\frac{V_G}{R_{TH} + R_L} \right)^2 R_L$$

$$\frac{dP_L}{dR_L} = 0$$

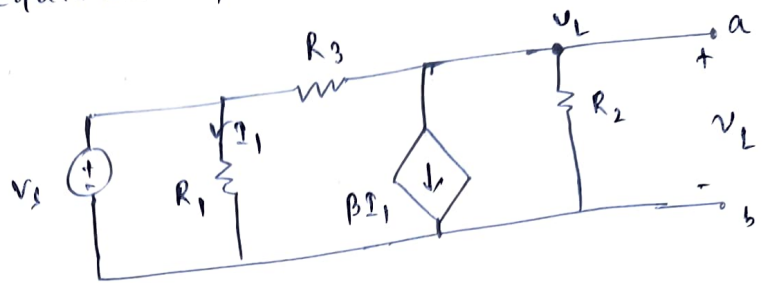
$$\Rightarrow \text{We get, } \boxed{R_{TH} = R_L}$$



⑧

Given $R_1 = 10\text{K}\Omega$, $R_2 = 75\text{K}\Omega$, $R_3 = 150\text{K}\Omega$, $\beta = 100$

a) For Thevenin Equivalent, remove the load resistance



so, applying nodal analysis at node V_L

$$\frac{V_L}{R_2} + \beta I_1 + \frac{V_L - V_s}{R_3} = 0 \quad \text{--- (1)}$$

Also, I_1 can be written as $I_1 = \frac{V_s}{R_1}$

so,

$$\text{Eqn (1)} \Rightarrow \frac{V_L}{R_2} + \beta \frac{V_s}{R_1} + \frac{V_L - V_s}{R_3} = 0$$

$$V_L \left(\frac{1}{R_2} + \frac{1}{R_3} \right) = V_s \left(\frac{\beta}{R_1} - \frac{1}{R_3} \right)$$

$$V_L \left(\frac{R_2 + R_3}{R_2 R_3} \right) = V_s \left(\frac{R_1 - \beta R_3}{R_1 R_3} \right)$$

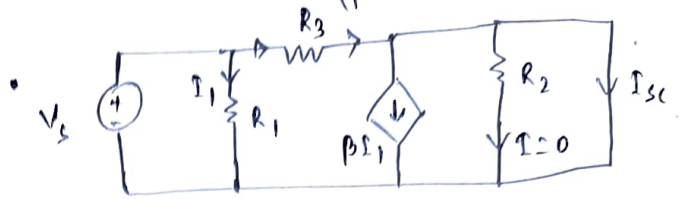
so, putting the value of R_2, R_3, R_1 & β

$$3V_L = -1499V_s$$

$$V_L = \frac{-1499V_s}{3}$$

$$[V_L = V_{th} = V_{oc}]$$

Now, finding the I_{sc} through load terminal $(\beta I_1 + I_{sc})$



so, Applying KVL :- $R_3 (\beta I_1 + I_{sc}) = V_s$ — (2)

$$I_1 = \frac{V_s}{R_1}$$

so, (2) $\Rightarrow \frac{V_s}{R_3} = \beta \frac{V_s}{R_1} + I_{sc}$

$$I_{sc} = V_s \left(\frac{1}{R_3} - \frac{\beta}{R_1} \right)$$

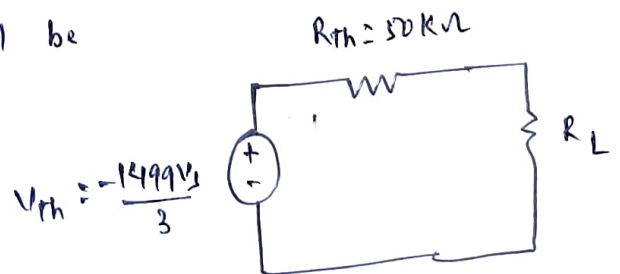
$$I_{sc} = V_s \left(\frac{R_1 - \beta R_3}{R_1 R_3} \right) \quad \left[\text{put } R_1, \beta, R_3 \text{ value} \right]$$

$$I_{sc} = -9.99 \times 10^{-3} V_s$$

$$\text{so, } R_{th} = \frac{V_L}{I_{sc}} = \frac{-1499 V_s}{3 \times (-9.99 \times 10^{-3}) V_s}$$

$$R_{th} = 50 \text{ k}\Omega$$

so, Thevenin equivalent will be



so, voltage across R_L

$$V_L = \frac{V_{th} \cdot R_L}{R_L + R_{th}}$$

$$\text{so, } V_L = \frac{-1499 V_s}{3} \cdot \frac{R_L}{R_L + 50000}$$

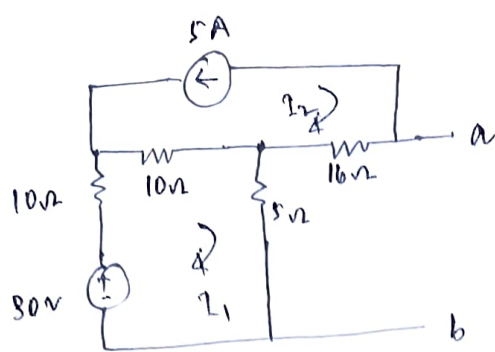
$$\text{so, } \boxed{\frac{V_L}{V_s} \approx \frac{-500 R_L}{R_L + 50000}}$$

⑧ When $R_L = 20000 \Omega$

$$\frac{V_L}{V_s} = \frac{-500 \times \cancel{20000}}{\cancel{70000}}$$

$$\boxed{\frac{V_L}{V_s} = -142.86}$$

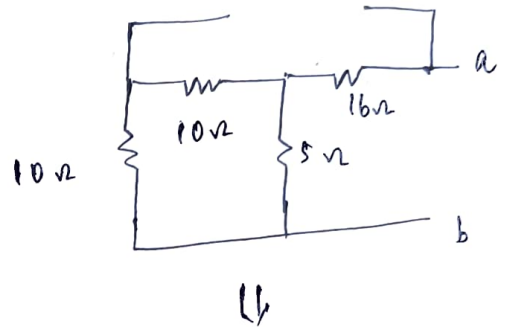
9



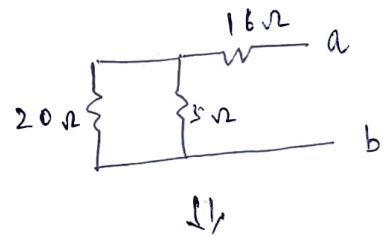
To find R_{ab} , deenergize the independent source i.e. short ckt the voltage source & open ckt the current source. so, the circuit will look like

~~R_{ab}~~ =

both 10Ω resistance are connected in series $\Rightarrow$



20Ω and 5Ω resistance are connected in parallel



so, $R_{ab} = R_{th} = R_N = 20\Omega$



Now, applying KVL in loop 1

$$30 = 10I_1 + 10(I_1 + 5) + 5I_1 \quad \left[\because I_1 + 5 = I_1 - I_2 \right]$$

$$30 = 25I_1 + 50$$

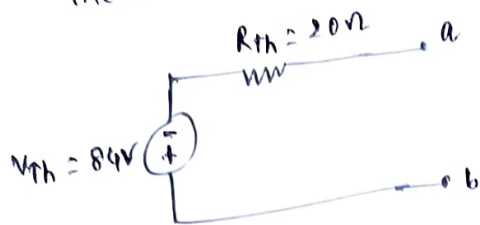
$$I_1 = -4/5 \text{ A}$$

A $I_2 = -5 \text{ A}$ from loop 2]

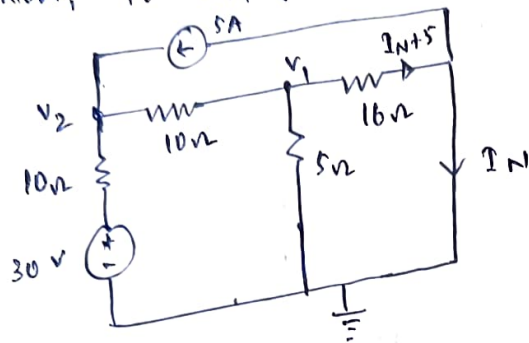
$$\text{so, } V_{ab} = 16I_2 + 5I_1$$

$$V_{ab} = -80 - 4 = -84 \text{ V}$$

So, Thevenin equivalent will look like



Now, to apply Norton Theorem, short circuit the a-b terminal



Applying Nodal analysis at node V_1

$$\frac{V_1}{5} + \frac{V_1 - V_2}{10} + \frac{V_1}{16} = 0$$

$$16V_1 + 8V_1 - 8V_2 + 5V_1 = 0$$

$$29V_1 - 8V_2 = 0 \quad \text{--- (A)}$$

again, Applying Nodal analysis at node V_2

$$-5 + \frac{V_2 - 30}{10} + \frac{V_2 - V_1}{10} = 0$$

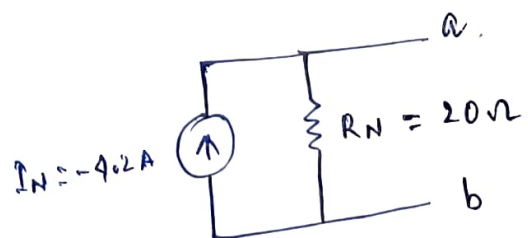
$$-50 + V_2 - 30 + V_2 - V_1 = 0$$

$$-V_1 + 2V_2 = 80 \quad \text{--- (B)}$$

solving Eqⁿ (A) & (B), we get $V_1 = \frac{64}{5} \text{ V}$, & $V_2 = \frac{232}{5} \text{ V}$

$$\text{So, } I_N + 5 = \frac{V_1}{16}$$

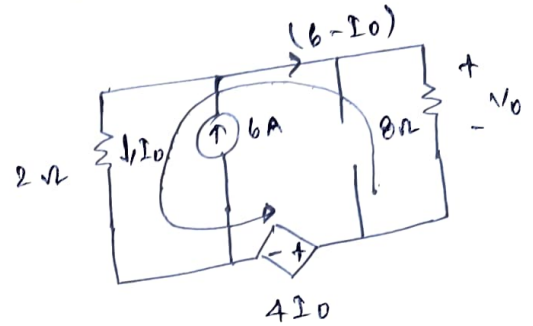
$$\Rightarrow I_N = -4.2 \text{ A}$$



10) Assume 6A source is acting alone in the ckt, then

so, voltage across 8Ω resistor

$$V_0 = 8(6 - I_0) \quad \text{--- (A)}$$



Apply KVL in the ckt

$$V_0 = 2I_0 - 4I_0$$

$$V_0 = -2I_0 \quad \text{--- (B)}$$

From eqn (A) & (B)

$$-2I_0 = 8(6 - I_0)$$

$$I_0 = 8A$$

$$(A) \Rightarrow V_0 = -16V$$

Now, assume 4A current source is acting alone

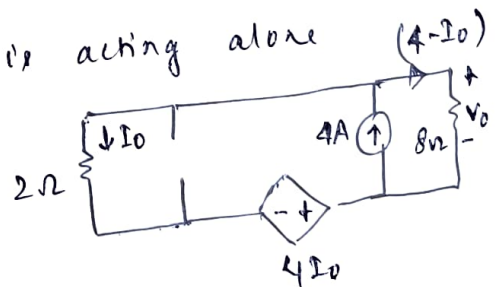
$$V_0 = 8(4 - I_0) \quad \text{--- (A)}$$

$$\text{KVL: } V_0 = -2I_0 \quad \text{--- (B)}$$

From eqn (A) & (B)

$$I_0 = \frac{16}{3} A$$

$$(A) \Rightarrow V_0 = -\frac{32}{3} V$$



so, A/c to superposition Theorem $V_{\text{net}} = V_0 = \left(-16 - \frac{32}{3}\right) V$
 $= -26.67 V$